

FORTIGATE NSE4

Lab 8 — SD-WAN

1. What SD-WAN Solves

A router with two WAN links and static routes can survive a link going down. It cannot survive a link that stays up but stops working — high latency, jitter, or packet loss. The route remains valid, traffic keeps using the degraded path, and the users notice before the monitoring does.

SD-WAN replaces the binary up/down decision with a measured one. It continuously probes each WAN link, compares the measurements against defined thresholds, and steers traffic to a path that meets them. It also collapses the configuration: instead of one route and one policy per WAN link, there is a single virtual interface holding all of them.

► 1.1 The Four Components

Component	Purpose
Zone	The virtual interface that groups the WAN links. Routes and firewall policies reference the zone, not the physical ports
Members	The physical interfaces participating, each with its next-hop gateway
Performance SLA	The health check — what to probe, how often, and the latency, jitter and packet loss thresholds that define an acceptable link
Rules	Which traffic uses which member, and by what strategy

2. Topology

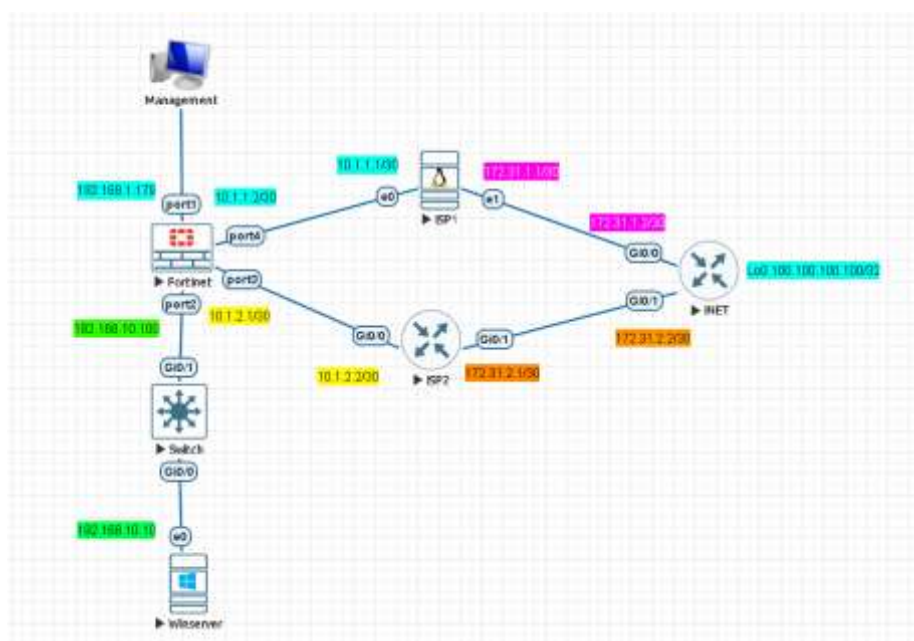


Figure 1 — Lab topology — two WAN paths to a common internet router

Link	FortiGate	Far end	Notes
WAN1	port4 10.1.1.2/30	ISP1 10.1.1.1/30	Linux — the degradable path
WAN2	port3 10.1.2.1/30	ISP2 10.1.2.2/30	vIOS router
ISP1 – INET	—	172.31.1.0/30	—
ISP2 – INET	—	172.31.2.0/30	—
LAN	port2 192.168.10.100/24	Winserver .10	Test client
Probe target	—	100.100.100.100/32	INET loopback

► 2.1 Why ISP1 Is a Linux Host

A router can only be switched off. Linux can be made *slow* — the `tc netem` queueing discipline injects controlled latency, jitter or packet loss on an interface while it continues forwarding normally.

That distinction is the point of the lab. Shutting a link tests failover, which a static route with link monitoring can also do. **Degrading a link tests SLA-based steering, which only SD-WAN can do.**

► 2.2 The Loopback as Probe Target

The health check measures the path to 100.100.100.100, a loopback on INET reachable through both ISPs. Because a loopback has no physical interface, it stays up regardless of which link is used — so a probe failure always means the *path* failed, never the target.

Each ISP path needs a return route on INET for this to work in both directions.

3. Configuring ISP1

The Linux host is configured as a router: two addressed interfaces, IP forwarding enabled, and a default route toward INET.

```
eve@ubuntu:~$ FGT_Inter=eth0
eve@ubuntu:~$ INET_IF=eth1
```

Figure 2 — Interface names stored in shell variables for readability

```
eve@ubuntu:~$ sudo ip addr add 10.1.1.1/30 dev $FGT_Inter
[sudo] password for eve:
eve@ubuntu:~$ sudo ip addr add 172.31.1.1/30 dev $INET_IF
eve@ubuntu:~$ sudo ip link set $FGT-Inter up
Cannot find device "-Inter"
eve@ubuntu:~$ sudo ip link set $FGT_Inter up
eve@ubuntu:~$ sudo ip link set $INET_IF up
eve@ubuntu:~$ sudo sysctl -w net.ipv4.ip_forward=1
```

Figure 3 — Addressing both interfaces, bringing them up, and enabling IP forwarding

```
sudo ip addr add 10.1.1.1/30 dev eth0      # toward FortiGate port4
sudo ip addr add 172.31.1.1/30 dev eth1    # toward INET
sudo ip link set eth0 up
sudo ip link set eth1 up
sudo sysctl -w net.ipv4.ip_forward=1
sudo ip route add default via 172.31.1.2
```

► 3.1 A Variable Typo That Produced Three Unrelated Symptoms

The output shows `Cannot find device '-Inter'` — the shell variable did not expand to the intended interface name, so the address was never applied.

The consequence appeared later and looked unrelated: adding the default route failed with `Next hop has invalid gateway`, because Linux had no interface in `172.31.1.0/30` and therefore no way to reach the next hop. The same missing address also meant INET could not ping ISP1 from its loopback.

One fault, three symptoms, none of which named the actual cause. Using literal interface names rather than variables removed all three at once.

These commands do not persist across a reboot. If ISP1 restarts mid-lab, the WAN1 path fails silently and the result is easily misread as an SD-WAN failover.

4. Configuring ISP2 and INET

```
Router(config)#hostname ISP2
ISP2(config)#interface Gi0/0
ISP2(config-if)#ip address 10.1.2.2 255.255.255.252
ISP2(config-if)#no shutdown
ISP2(config-if)#exit
ISP2(config)#int
*Sep 22 13:06:26.832: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed
e to up
*Sep 22 13:06:27.836: %LINEPROTO-5-UPDOWN: Line protocol on Interface Gigab
ernet0/0, changed state to uperface Gi0/1
ISP2(config-if)#ip add
ISP2(config-if)#ip address 172.31.2.1 255.255.255.252
ISP2(config-if)#no sh
ISP2(config-if)#no shutdown
ISP2(config-if)#ip route 0.0
*Sep 22 13:06:43.661: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed
e to up
*Sep 22 13:06:44.662: %LINEPROTO-5-UPDOWN: Line protocol on Interface Gigab
ernet0/1, changed state to up.0.0
ISP2(config-if)#ip route 0.0.0.0 0.0.0.0 172.31.2.2
```

Figure 4 — ISP2 interface addressing and default route toward INET

```
hostname INET
interface Gi0/0
  ip address 172.31.1.2 255.255.255.252
  no shutdown
interface Gi0/1
  ip address 172.31.2.2 255.255.255.252
  no shutdown
interface Loopback0
  ip address 100.100.100.100 255.255.255.255
ip route 10.1.1.0 255.255.255.252 172.31.1.1
ip route 10.1.2.0 255.255.255.252 172.31.2.1
```

INET configuration — the two static routes are the return paths for each WAN link

► 4.1 Validating Before the FortiGate

```
INET#ping 10.1.2.2 source Loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.2, timeout is 2 seconds:
Packet sent with a source address of 100.100.100.100
!!!!
```

Figure 5 — INET reaching the WAN link from its loopback source

A plain ping from INET would use the outgoing interface address, which succeeds trivially because the link is directly connected. Sourcing from `Loopback0` forces the reply to route back through the return route — the exact path the FortiGate's health check will depend on.

Run this for both ISPs before creating any SD-WAN object. It eliminates the most common cause of a member that reports dead from the moment it is created.

5. FortiGate Interfaces

Figure 6 — WAN interface configuration with role set to WAN

Interface	Alias	Address	Role	Access
port4	WAN1-ISP1	10.1.1.2/30	WAN	PING
port3	WAN-ISP-2	10.1.2.1/30	WAN	PING

Administrative access is limited to PING, which the health check requires and nothing else. Setting the role to WAN affects how the interface is presented in the GUI and which features are offered; it does not change forwarding.

Clear existing default routes first. An interface already referenced by a static route cannot be added as an SD-WAN member. More subtly, **a leftover default route via another interface at the same distance creates equal-cost paths alongside the SD-WAN route**, splitting traffic unpredictably and making every subsequent test unreadable.

6. SD-WAN Members

Figure 7 — Both members assigned to the virtual-wan-link zone with their next-hop gateways

Member	Interface	Gateway	Cost	Priority
1	port4 (WAN1-ISP1)	10.1.1.1	0	1
2	port3 (WAN-ISP-2)	10.1.2.2	0	1

► 6.1 Priority Here Is Not Path Preference

The Priority field on a member is route priority — it influences the routing table, not SLA selection. Both members are left at the same value deliberately.

Which member carries traffic is decided by the rule's interface preference order and its chosen strategy, evaluated after the health check. Setting different priorities here would bias routing before SD-WAN logic runs and obscure what the rule is doing.

7. Performance SLA

Figure 8 — The health check definition and its thresholds

Setting	Value	Meaning
Probe mode / Protocol	Active / Ping	The FortiGate generates its own probes rather than inferring quality from user traffic
Server	100.100.100.100	The INET loopback
Participants	Both members	Each member is measured independently
Latency threshold	100 ms	—
Jitter threshold	30 ms	—

Setting	Value	Meaning
Packet loss threshold	2 %	—
Check interval	500 ms	Probe frequency
Failures before inactive	3	About 1.5 seconds before a member is declared dead
Restore link after	5	Five consecutive good probes before a recovered member is trusted again

► 7.1 Why Recovery Is Slower Than Failure

Three failures mark a link dead; five successes bring it back. **The asymmetry is deliberate — it prevents a marginal link from flapping traffic back and forth as it hovers around a threshold.**

Worth knowing during testing: after removing an impairment, traffic does not return instantly, and that delay is correct behaviour rather than a fault.

8. Route, Policy and Rule

► 8.1 Static Route

The screenshot shows the 'New Static Route' configuration window. It has a title bar 'New Static Route'. Below it, there are two tabs: 'Subnet' (selected) and 'Internet Service'. The 'Destination' field is set to '0.0.0.0/0.0.0.0'. The 'Interface' field is set to 'virtual-wan-link'. There is a 'Comments' field with the placeholder text 'Write a comment...'. The 'Status' is set to 'Enabled' (indicated by a green circle with a plus sign) and 'Disabled' (indicated by a red circle with a minus sign) is also visible.

Figure 9 — A single default route pointing at the SD-WAN zone

One route replaces what would otherwise be one per WAN link. The zone resolves to whichever member the rules select, so the routing table does not change when a path is switched.

► 8.2 Firewall Policy



Figure 10 — LAN to SD-WAN zone policy with NAT enabled

The outgoing interface is the zone, so one policy covers both WAN links. NAT is enabled here — unlike the inter-VLAN policies in earlier labs — because the private LAN addressing is not routable beyond the FortiGate.

► 8.3 SD-WAN Rule

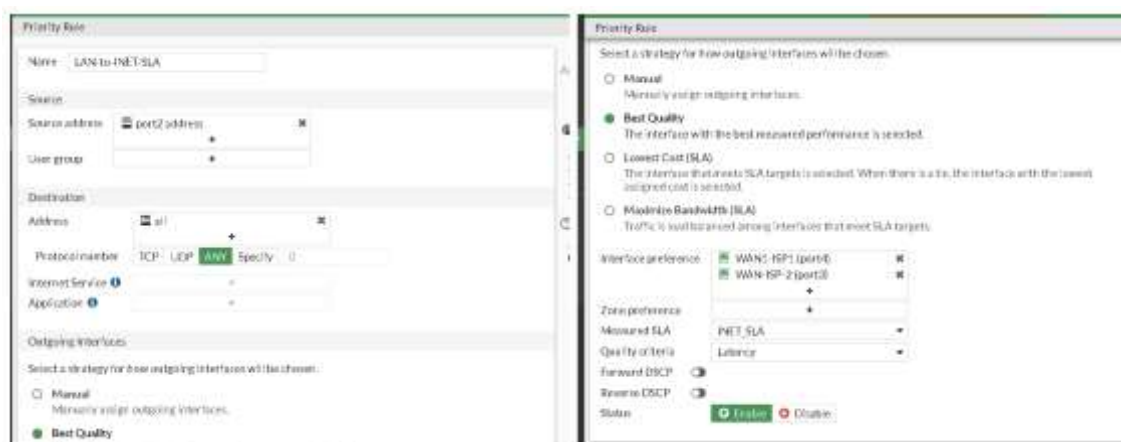


Figure 11 — The rule: source, destination, strategy and interface preference

Setting	Value
Source / Destination	port2 address / all
Strategy	Best Quality
Interface preference	WAN1-ISP1 (port4), then WAN-ISP-2 (port3)
Measured SLA	INET_SLA
Quality criteria	Latency

► 8.4 What Best Quality Compares — and What It Does Not

Best Quality selects the member with the best measurement of the **single** chosen quality criterion — here, latency. **The jitter and packet loss thresholds defined in the SLA are still measured and displayed, but they do not influence selection in this mode.**

Lowest Cost (SLA) behaves differently: it honours the interface preference order deterministically and switches when *any* of the three thresholds is breached. For a deployment where jitter or loss matters — voice, or industrial protocols sensitive to gaps — that is the correct strategy.

Best Quality was used here because the test being demonstrated is latency-based, which is the one criterion it evaluates. The trade-off is noted rather than hidden.

9. A Setting the GUI Does Not Expose

```
FortiGate-VM64-KVM # config system global
FortiGate-VM64-KVM (global) # set snat-route-change enable
FortiGate-VM64-KVM (global) # end
FortiGate-VM64-KVM #
```

Figure 12 — Enabling snat-route-change from the CLI

```
config system global
    set snat-route-change enable
end
```

By default, an established session that has been source-NATed keeps its original egress path even after SD-WAN selects a different member. New sessions follow the new path; existing ones do not.

Why this is required for the test to work: a continuous ping is a single long-lived flow. Without this setting it would remain pinned to the degraded link, the latency would climb, traffic would never move, and the conclusion would be that SLA steering does not work — when in fact it had already reselected. **The configuration is correct, the test appears to fail, and nothing in the GUI indicates why.**

10. Baseline

```
FortiGate-VM64-KVM # diagnose sys sdwan health-check
Health Check(INET_SLA):
Seq(1 port3): state(alive), packet-loss(0.000%), latency(4.975), jitter(1.011), bandwidth-up(9999999), bandwidth-dw(9999999), bandwidth-b
i(19999998) sla_map=0x1
Seq(2 port4): state(alive), packet-loss(0.000%), latency(4.360), jitter(1.485), bandwidth-up(9999998), bandwidth-dw(9999999), bandwidth-b
i(19999997) sla_map=0x1
```

Figure 13 — Both members alive and within SLA before any impairment

Both links report roughly 4 to 5 ms latency, around 1.5 ms jitter, and no packet loss. The `sla_map=0x1` value confirms each member is meeting SLA target 1.

```
Service(1): Address Mode(IPV4) flags=0x200 use-shortcut-sla
  Gen(1), TOS(0x0/0x0), Protocol(0: 1->65535), Mode(priority), link-cost-factor(latency), link-cost-threshold(10), health-check(INET_SLA)
  Members(2):
    1: Seq_num(2 port4), alive, latency: 4.382, selected
    2: Seq_num(1 port3), alive, latency: 4.954, selected
  Src address(1):
    192.168.10.0-192.168.10.255
  Dst address(1):
    0.0.0.0-255.255.255.255
```

Figure 14 — The service rule showing both members alive, port4 selected

```
C:\Users\Administrator>tracert -d 100.100.100.100

Tracing route to 100.100.100.100 over a maximum of 30 hops

  1      1 ms      2 ms      1 ms    192.168.10.100
  2      4 ms      2 ms      2 ms    10.1.1.1
  3      5 ms      3 ms      2 ms    100.100.100.100

Trace complete.
```

Figure 15 — Traceroute from the LAN client — hop 2 is 10.1.1.1, the ISP1 path

Traceroute is the strongest evidence in this lab. The health-check and service outputs show what the FortiGate has *decided*. A traceroute from the client shows what the packets actually *did*. Hop 2 names the ISP carrying the traffic — 10.1.1.1 for ISP1, 10.1.2.2 for ISP2 — so a before-and-after pair proves the forwarding path changed, not merely a status field.

11. Latency Test

A continuous ping was started from the LAN client, then 200 ms of delay was injected on the ISP1 interface facing the FortiGate.

```
eve@ubuntu:~$ sudo tc qdisc add dev eth0 root netem delay 200ms
```

Figure 16 — Injecting 200 ms of latency with `tc netem`

```
sudo tc qdisc add dev eth0 root netem delay 200ms
```

The link remains up and continues forwarding throughout. **Only its quality changes — which is precisely the condition a link-state failover cannot detect.**

```

FortiGate-VM04-KVM # diagnose sys sdean health-check
Health Check(INET.SLA):
Seq(1 port3): state(alive), packet-loss(0.000%) latency(3.515), jitter(1.446), bandwidth-up(9999998), bandwidth-dw(9999999), bandwidth-bi(19999997) sla_map=0x1
Seq(2 port4): state(alive), packet-loss(0.000%) latency(212.327), jitter(3.600), bandwidth-up(9999998), bandwidth-dw(9999999), bandwidth-bi(19999997) sla_map=0x0

```

Figure 17 — Health check after impairment — port4 latency has risen to 212 ms

Latency on port4 climbs from about 4 ms to 212 ms, past the 100 ms threshold, while packet loss stays at zero and the member remains `alive`. The link is healthy by any up/down measure and unusable by the SLA's.

```

Reply from 100.100.100.100: bytes=32 time=8ms TTL=253
Reply from 100.100.100.100: bytes=32 time=3ms TTL=253
Reply from 100.100.100.100: bytes=32 time=5ms TTL=253
Request timed out.
Reply from 100.100.100.100: bytes=32 time=4ms TTL=253
Reply from 100.100.100.100: bytes=32 time=3ms TTL=253
Reply from 100.100.100.100: bytes=32 time=7ms TTL=253

```

Figure 18 — The continuous ping across the transition — a single lost packet

```

C:\Users\Administrator>tracert -d 100.100.100.100

Tracing route to 100.100.100.100 over a maximum of 30 hops

  1    1 ms    1 ms    1 ms  192.168.10.100
  2    5 ms    3 ms    2 ms  10.1.2.2
  3    6 ms    4 ms    3 ms  100.100.100.100

Trace complete.

```

Figure 19 — Traceroute after impairment — hop 2 is now 10.1.2.2, the ISP2 path

The result: one packet lost, then service continues. The client never changed configuration, the route table still holds a single default route, and the degraded link was never down. **A static route with link monitoring would have kept every packet on the 212 ms path, because by its measure nothing was wrong.**

12. Recovery

```

eve@ubuntu:~$ sudo tc qdisc del dev eth0 root

```

Figure 20 — Removing the impairment

```
sudo tc qdisc del dev eth0 root
```

```

Tracing route to 100.100.100.100 over a maximum of 30 hops

  1    1 ms    3 ms    1 ms  192.168.10.100
  2    3 ms    3 ms    2 ms  10.1.2.2
  3    *      5 ms    4 ms  100.100.100.100

Trace complete.

C:\Users\Administrator>tracert -d 100.100.100.100

Tracing route to 100.100.100.100 over a maximum of 30 hops

  1    2 ms    1 ms    1 ms  192.168.10.100
  2    2 ms    4 ms    2 ms  10.1.1.1
  3    4 ms    5 ms    3 ms  100.100.100.100

Trace complete.

```

Figure 21 — Two consecutive traceroutes during recovery: still on ISP2, then back on ISP1

The first traceroute after removing the delay still shows hop 2 as 10.1.2.2 — the ISP2 path — because the recovered member must pass five consecutive probes before it is trusted again. The second traceroute, moments later, shows 10.1.1.1: traffic has returned to ISP1.

The transition is visible, not instant. Capturing both traceroutes rather than only the final state is what demonstrates the recovery timer doing its job. An immediate switch back would indicate the damping was not configured.

► 12.1 Confirmation from the Traffic Log

Source	Device	Destination	Application Name	Result	Policy ID	Destination Interface
192.168.10.10	WIN-LIP2GAPKSGGQ	100.100.100.100	PING	✓ 1.14 kB / 1.14 kB	Test-SD-WAN(4)	WAN1-ISP1 (port4)
Administrator (192.168.10.10)	WIN-LIP2GAPKSGGQ	100.100.100.100	PING	✓ 900 B / 900 B	Test-SD-WAN(4)	WAN-ISP-2 (port3)

Figure 22 — Forward traffic log showing sessions on both WAN interfaces

Two sessions from the same client to the same destination, matched by the same policy, leaving through different interfaces — WAN1-ISP1 (port4) and WAN-ISP-2 (port3). This is the independent record of the path change, produced by the firewall's own logging rather than by a diagnostic command.

13. Summary

Objective	Result
Aggregate two WAN links into one logical interface	Achieved — one zone, one static route, one policy
Measure path quality continuously	Achieved — both members reporting latency, jitter and loss against thresholds
Steer traffic away from a degraded but functional link	Achieved — 200 ms injected latency moved traffic with one packet lost
Return traffic when the link recovers	Achieved — restored after the recovery timer, visible across two traceroutes
Verify independently of the FortiGate's own status output	Achieved — client traceroutes and the forward traffic log both confirm the path change